

# Support-Ticket Lifecycle — Optimisation Recommendation

Week 12 · InGen fleet support (Fari, Sentinel, Aido) · simulated workflow on the Week 7 warehouse · prepared by Ziheng Wang

## Recommendation in one line

**Stock parts regionally before you hire.**

A local parts buffer in Europe and APAC cuts mean cycle time **23%** (26.0h → 20.0h). Adding a Field Ops FTE cuts it **8%**. Doing both cuts it **29%**. Rerouting Tier-1 triage — the obvious lever — changes **nothing measurable**.

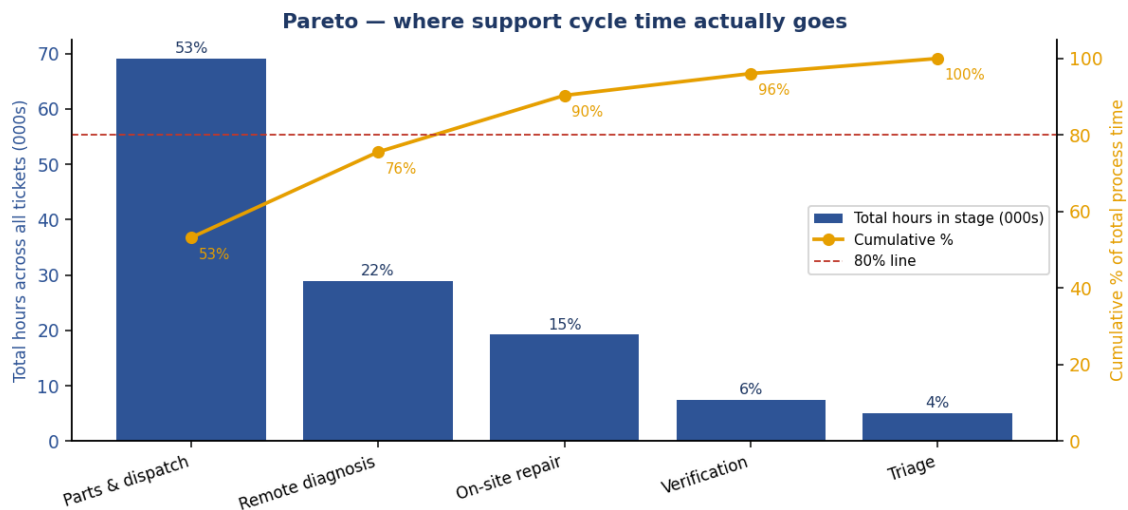
## The process

Intake → Triage (Tier 1) → Remote diagnosis (Tier 2) → [ *Parts & dispatch* → *On-site repair (Field Ops)* ] → Verification & closure. The bracketed path runs only when a ticket needs a physical part or site visit — 37% of tickets. Every one of the 5,000 Week 7 tickets was replayed through this lifecycle; stage durations reconcile exactly to each ticket's total cycle time, and every ticket\_key joins 1:1 back to fact\_support\_tickets.

## Where the time goes — top 3 stages are 90% of it

| Stage            | % of all process time | Cumulative | Of which waiting | Median | P90   |
|------------------|-----------------------|------------|------------------|--------|-------|
| Parts & dispatch | 53.2%                 | 53.2%      | 100%             | 23.9h  | 77.5h |
| Remote diagnosis | 22.3%                 | 75.5%      | 31%              | 3.9h   | 12.2h |
| On-site repair   | 14.8%                 | 90.3%      | 48%              | 7.1h   | 21.0h |
| Verification     | 5.7%                  | 96.1%      | 27%              | 1.2h   | 2.7h  |
| Triage           | 3.9%                  | 100.0%     | 47%              | 0.7h   | 2.1h  |

**Parts & dispatch alone is 53% of all process time and is 100% waiting** — nobody is working during it, so no amount of headcount shortens it. On-site repair is 48% queueing, which headcount *can* shorten. That distinction drives the whole recommendation.



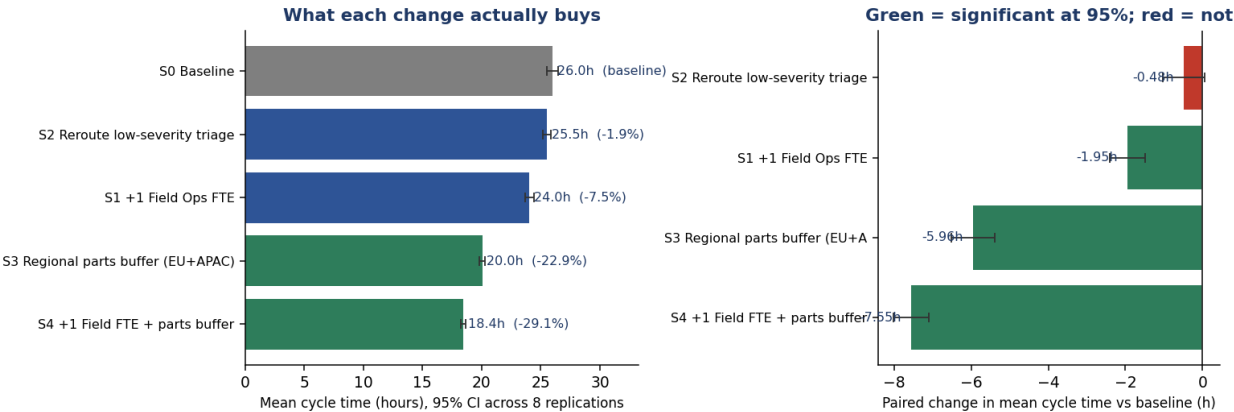
# What moves cycle time — and what doesn't

Regression of log(cycle time) on the drivers in the brief (product, severity, geography, weekday, team workload), n = 5,000, 95% confidence intervals.

| Driver                                   | Effect on cycle time | 95% CI           | Significant? |
|------------------------------------------|----------------------|------------------|--------------|
| severity: Critical                       | -39.9%               | [-46.6%, -32.3%] | yes          |
| severity: High                           | -19.9%               | [-26.1%, -13.2%] | yes          |
| severity: Low                            | +27.8%               | [+18.5%, +37.8%] | yes          |
| Tier-1 queue length (per ticket waiting) | +10.3%               | [+6.5%, +14.3%]  | yes          |
| region: APAC                             | +7.1%                | [-0.6%, +15.5%]  | no           |
| region: Europe                           | +5.3%                | [-2.1%, +13.3%]  | no           |
| product: Aido Humanoid                   | +4.4%                | [-5.4%, +15.3%]  | no           |
| weekday: Friday                          | -8.2%                | [-18.2%, +3.0%]  | no           |

**These drivers together explain only 4% of the variance in cycle time.** Add one structural fact — did the ticket need a part? — and R<sup>2</sup> jumps to **72%**: a dispatched ticket takes **+578%** longer [+555%, +601%]. Severity and queue length are real and significant (a Critical ticket clears 40% faster; each ticket already queued adds +10.3%), but they are second-order. **Support cycle time is not a people-management problem; it is a parts problem.** Weekday shows no effect, which is expected — this model runs follow-the-sun cover with no shift calendar (see limitations).

## Simulated changes — 8 replications each, paired to the same seeds



| Scenario                           | Mean cycle | Δ vs baseline | 95% CI of Δ      | Verdict              |
|------------------------------------|------------|---------------|------------------|----------------------|
| S0 Baseline                        | 26.0h      | —             | —                | reference            |
| S2 Reroute low-severity triage     | 25.5h      | -1.9%         | [-1.02h, 0.05h]  | no measurable effect |
| S1 +1 Field Ops FTE                | 24.0h      | -7.5%         | [-2.41h, -1.50h] | significant          |
| S3 Regional parts buffer (EU+APAC) | 20.0h      | -22.9%        | [-6.52h, -5.39h] | significant          |
| S4 +1 Field FTE + parts buffer     | 18.4h      | -29.1%        | [-8.02h, -7.09h] | significant          |

# Recommendation, trade-offs and limits

## Recommendation

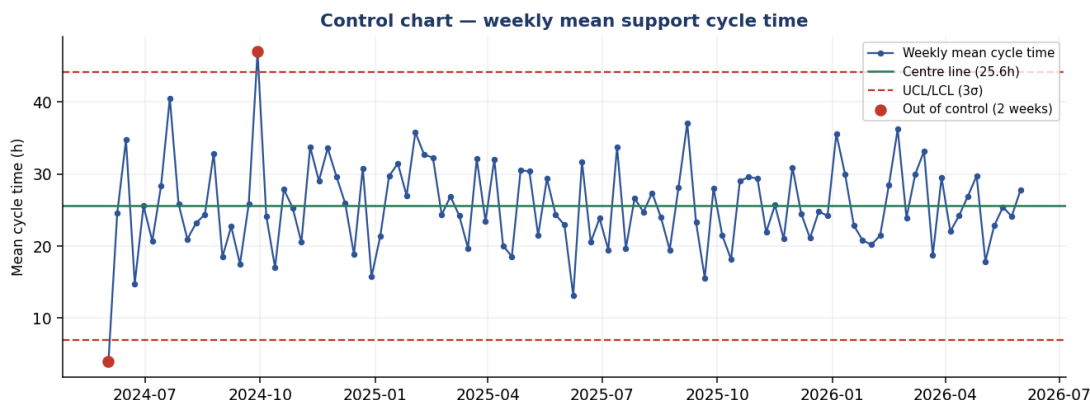
- 1. Fund a regional parts buffer in Europe and APAC first.** It is the single largest lever in the model: **23%** off mean cycle time [-6.5h, -5.4h], and it also pulls P90 down from 64h to 47h — the tail is what customers actually complain about.
  - 2. Add the Field Ops FTE second.** Worth a real **8%**, and it nearly eliminates field-queue wait (4.9h → 0.4h). But its ceiling is set by the stage's share of the process — On-site repair is only ~15% of total time, so perfect staffing there cannot buy more than ~15%.
  - 3. Do not spend effort rerouting Tier-1 triage.** Triage is 4% of process time; the simulated change is -1.9% with a CI spanning zero — statistically indistinguishable from doing nothing.
- Together (1+2): 29% off mean cycle time.**

## Trade-offs, named

| Lever                 | Cost                                                                                   | Cycle time              | Service level                              | Risk                                            |
|-----------------------|----------------------------------------------------------------------------------------|-------------------------|--------------------------------------------|-------------------------------------------------|
| Regional parts buffer | Working capital tied up in inventory at two sites; obsolescence risk as hardware revs. | -23% mean; P90 64h→47h  | Biggest tail improvement — best CSAT lever | Stock the wrong SKUs and the gain evaporates    |
| +1 Field Ops FTE      | Recurring salary; hiring lead time; needs utilisation to stay justified.               | -8% mean                | Removes field queue almost entirely        | Capped by the stage's 15% share of process time |
| Reroute Tier-1 triage | Engineering time to build auto-routing/self-service.                                   | -1.9% — not significant | No measurable gain                         | Opportunity cost of building it                 |

## Is the process in control?

Weekly mean cycle time over 105 weeks sits on a centre line of 25.6h with  $3\sigma$  limits at 6.9–44.2h; **2 weeks breach the upper limit**. The process is broadly in statistical control — meaning the long cycle times are not an anomaly to be firefighted, they are what this process delivers by design. Fixing them requires changing the process (parts availability), not chasing individual bad weeks.



## Limitations — read before acting

**This runs on synthetic data.** The Week 7 warehouse carries no stage, team or workload fields, so the lifecycle layer was generated by a documented model (src/process/process\_model.py) replaying Week 7's real ticket stream. Every number above is therefore a property of that model, **not a measurement of InGen's support org**. Team sizes (Tier 1 = 1, Tier 2 = 2, Field Ops = 1) were chosen to load the teams realistically against the observed ~6.8 tickets/day, not observed from data. **What does transfer** is the method and the pipeline: point the same code at real ticket data with stage timestamps and it produces the same diagnostic. The driver regression was validated against the model's known generating parameters and recovered **7/9** effect directions — the two misses are small, non-significant product effects that act only on a stage covering 15% of time for 37% of tickets, so they are diluted below noise. No shift calendar is modelled, so the null weekday result should not be read as evidence that weekday doesn't matter in reality.

**The ask:** real support-ticket data with stage-level timestamps would turn this from a validated method into an actual answer.

Reproduce: `python -m src.process.stage_generator → cycle_time → drivers → capacity_sim → build_report`. Seeded (SEED=20260710); 8 replications per scenario; paired deltas against identical seeds. Ziheng Wang · inGen Data Analyst internship · Week 12.